

## UENO, Japan

WMO: 476490

Lat: 34.7617N

Lon: 136.1419E

Elev: 528

StdP: 14.42

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 26.6       | 28.2 | 15.6                          | 12.4 | 39.0 | 18.6 | 14.3 | 38.6 | 22.2                  | 37.1 | 20.3 | 37.8 | 3.5                   | 160  | 0.406 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 16.4                   | 93.5            | 76.5 | 91.4 | 76.1 | 89.1 | 75.4 | 78.6                | 88.1 | 77.7 | 86.8 | 76.9 | 85.4 | 8.6                  | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 76.5                            | 140.8 | 81.3 | 75.5 | 136.1 | 81.1 | 74.5 | 131.7 | 80.8 | 42.5          | 88.1 | 41.6 | 86.9 | 40.8 | 85.7 | 82.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 18.3              | 15.9 | 14.0 | DB                         | 22.8 | 96.5               | 2.0 | 1.6                                                | 21.4 | 97.7       | 20.2 | 98.6       | 19.1 | 99.5       | 17.7 | 100.6 |
|                   |      |      | WB                         | 21.6 | 80.0               | 2.0 | 1.2                                                | 20.2 | 80.9       | 19.0 | 81.6       | 17.9 | 82.2       | 16.4 | 83.1  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 59.1   | 38.5  | 39.7  | 45.5  | 55.3  | 64.8  | 71.7  | 79.6  | 81.4  | 74.0  | 63.0  | 51.8  | 42.6 |
|                                                                        | DBStd       | 15.66  | 3.63  | 4.63  | 5.36  | 5.97  | 4.55  | 4.33  | 4.02  | 3.34  | 5.20  | 5.32  | 5.57  | 4.63 |
|                                                                        | HDD50       | 1109   | 357   | 291   | 163   | 20    | 0     | 0     | 0     | 0     | 0     | 0     | 44    | 234  |
|                                                                        | HDD65       | 3687   | 821   | 709   | 606   | 295   | 60    | 2     | 0     | 0     | 3     | 102   | 396   | 693  |
|                                                                        | CDD50       | 4430   | 0     | 2     | 22    | 177   | 458   | 652   | 917   | 973   | 721   | 404   | 98    | 6    |
|                                                                        | CDD65       | 1533   | 0     | 0     | 0     | 3     | 52    | 204   | 452   | 508   | 274   | 40    | 0     | 0    |
|                                                                        | CDH74       | 12898  | 0     | 0     | 0     | 57    | 438   | 1243  | 4093  | 5063  | 1842  | 162   | 0     | 0    |
|                                                                        | CDH80       | 4745   | 0     | 0     | 0     | 5     | 79    | 322   | 1645  | 2137  | 542   | 15    | 0     | 0    |
|                                                                        | WSAvg       | 5.6    | 6.7   | 6.4   | 6.4   | 6.0   | 5.7   | 5.1   | 5.0   | 5.1   | 5.2   | 4.8   | 4.8   | 6.1  |
|                                                                        | PrecAvg     | 59.0   | 2.4   | 2.8   | 4.2   | 4.3   | 5.9   | 7.7   | 7.4   | 5.2   | 7.9   | 5.0   | 3.4   | 2.4  |
| (15) Precipitation                                                     | PrecMax     | 72.4   | 5.6   | 5.0   | 7.4   | 8.3   | 14.2  | 13.6  | 15.1  | 14.8  | 16.4  | 12.0  | 9.1   | 4.5  |
|                                                                        | PrecMin     | 40.1   | 0.7   | 0.9   | 1.9   | 1.3   | 2.1   | 3.0   | 1.6   | 1.3   | 1.5   | 0.8   | 0.4   | 0.3  |
|                                                                        | PrecStd     | 9.1    | 1.2   | 1.1   | 1.5   | 1.7   | 3.0   | 2.6   | 3.6   | 3.1   | 3.9   | 2.4   | 2.1   | 1.1  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |      |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 55.4   | 61.9  | 69.6  | 79.7  | 86.3  | 90.8  | 95.4  | 96.5  | 91.7  | 81.9  | 71.1  | 61.7  |      |
|                                                                        | 0.4% MCWB   | 47.6   | 52.0  | 53.6  | 60.2  | 64.4  | 74.1  | 76.9  | 77.0  | 75.8  | 70.2  | 60.5  | 54.9  |      |
|                                                                        | 2% DB       | 51.8   | 56.3  | 64.7  | 74.8  | 82.1  | 86.7  | 93.2  | 94.1  | 88.6  | 78.1  | 67.8  | 57.5  |      |
|                                                                        | 2% MCWB     | 44.4   | 47.3  | 51.5  | 57.5  | 63.4  | 71.3  | 76.5  | 76.7  | 74.3  | 66.9  | 58.2  | 50.3  |      |
|                                                                        | 5% DB       | 49.2   | 52.4  | 60.4  | 71.2  | 78.9  | 83.7  | 91.1  | 92.3  | 85.9  | 75.3  | 64.9  | 54.7  |      |
|                                                                        | 5% MCWB     | 42.5   | 44.8  | 50.3  | 56.6  | 62.2  | 69.3  | 76.2  | 76.4  | 73.2  | 64.9  | 57.1  | 48.3  |      |
|                                                                        | 10% DB      | 46.6   | 49.3  | 57.1  | 67.6  | 75.7  | 80.9  | 88.6  | 90.0  | 83.1  | 72.7  | 62.1  | 52.1  |      |
|                                                                        | 10% MCWB    | 40.7   | 43.2  | 48.8  | 55.7  | 61.5  | 68.8  | 75.6  | 76.0  | 71.9  | 63.6  | 55.9  | 46.1  |      |
|                                                                        | 0.4% WB     | 51.1   | 54.8  | 58.5  | 66.5  | 71.0  | 77.0  | 79.3  | 80.1  | 78.3  | 73.4  | 63.9  | 56.7  |      |
|                                                                        | 0.4% MCDB   | 53.4   | 60.1  | 62.3  | 71.2  | 77.2  | 84.8  | 89.8  | 89.5  | 85.9  | 78.3  | 67.1  | 59.4  |      |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 46.2   | 50.0  | 55.3  | 62.6  | 68.0  | 74.9  | 78.2  | 79.0  | 76.5  | 70.2  | 61.2  | 52.6  |      |
|                                                                        | 2% MCDB     | 49.4   | 53.2  | 60.9  | 68.9  | 74.2  | 82.0  | 88.2  | 88.4  | 83.8  | 75.2  | 64.6  | 55.9  |      |
|                                                                        | 5% WB       | 43.5   | 46.6  | 52.6  | 60.1  | 66.0  | 73.2  | 77.4  | 78.1  | 75.3  | 67.7  | 59.1  | 49.4  |      |
|                                                                        | 5% MCDB     | 48.0   | 50.4  | 58.0  | 66.8  | 73.2  | 79.8  | 86.8  | 87.2  | 82.3  | 72.4  | 63.1  | 53.1  |      |
|                                                                        | 10% WB      | 41.2   | 43.8  | 49.8  | 57.8  | 64.1  | 71.4  | 76.5  | 77.3  | 74.2  | 65.4  | 57.0  | 46.8  |      |
|                                                                        | 10% MCDB    | 45.6   | 48.5  | 55.5  | 64.4  | 71.5  | 77.5  | 85.2  | 86.0  | 80.4  | 70.5  | 61.3  | 51.2  |      |
| (35) Mean Daily Temperature Range                                      | MDBR        | 14.7   | 16.2  | 19.0  | 20.9  | 19.4  | 15.5  | 15.3  | 16.4  | 15.4  | 16.9  | 17.8  | 16.1  |      |
|                                                                        | 5% DB MCDBR | 19.9   | 23.2  | 26.3  | 27.5  | 26.4  | 20.9  | 19.3  | 20.0  | 18.2  | 19.7  | 20.8  | 20.5  |      |
|                                                                        | 5% DB MCWBR | 13.3   | 14.9  | 15.5  | 13.1  | 10.7  | 7.8   | 6.2   | 6.0   | 6.8   | 9.6   | 12.8  | 13.9  |      |
|                                                                        | 5% WB MCDBR | 17.8   | 18.8  | 20.4  | 19.5  | 18.3  | 15.5  | 16.8  | 17.1  | 15.0  | 14.8  | 16.3  | 16.9  |      |
|                                                                        | 5% WB MCWBR | 13.4   | 14.2  | 14.6  | 12.0  | 9.8   | 6.9   | 6.0   | 5.7   | 6.3   | 8.4   | 11.6  | 13.2  |      |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |      |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.399  | 0.438 | 0.497 | 0.534 | 0.540 | 0.559 | 0.559 | 0.527 | 0.487 | 0.450 | 0.413 | 0.394 |      |
|                                                                        | taud        | 2.145  | 2.052 | 1.915 | 1.875 | 1.925 | 1.950 | 2.002 | 2.081 | 2.162 | 2.183 | 2.192 | 2.174 |      |
|                                                                        | Ebn at Noon | 246    | 250   | 245   | 243   | 243   | 238   | 237   | 242   | 246   | 244   | 239   | 239   |      |
|                                                                        | Edh at Noon | 38     | 46    | 58    | 63    | 61    | 59    | 56    | 51    | 45    | 40    | 36    | 35    |      |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 739    | 949   | 1257  | 1512  | 1640  | 1473  | 1563  | 1606  | 1264  | 1019  | 798   | 670   |      |
|                                                                        | RadStd      | 50     | 75    | 90    | 144   | 176   | 139   | 210   | 143   | 113   | 104   | 71    | 45    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (37 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page